

On the design of paired comparison experiments with quantitative factors in a constrained design region

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Abstract

For a linear model of paired comparison experiments, we consider the problem of deriving optimal approximate designs under general Φ_q -criteria in a setting where the explanatory variables are quantitative factors with values in a constrained design region. We establish complete class results in order to reduce the set of candidate designs to a smaller class of designs with a simple structure. Based on these results, we derive optimal designs for the Φ_q -criteria that are equivalent to classical E -, A -, and D -optimality. We complement the results with efficiency comparisons for some designs from the reduced class, which can be readily implemented in practice.

Keywords: Paired comparison experiment, Optimal design, Quantitative factors, Constrained design region, Linear model

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1 Introduction

The method of paired comparisons is an experimental technique that is based on comparing objects, alternatives, or items in pairs. For each pair, a response is observed that reflects, in a certain sense, the superiority of one object over the other, often in terms of utility or preference. This approach is particularly suited to situations where direct evaluations of single objects may be difficult to obtain or unreliable.

Responses can either be continuous or categorical, implying the use of different models for analysis. In linear paired comparisons, the response variable is continuous and the relationship between the response and the explanatory variables is modelled by a general linear model. Prominent examples of paired comparison models with a categorical response include the Bradley-Terry model and certain types of discrete choice models. For additional background on the history, models, and applications of paired comparisons, we refer to [David \(1988\)](#), [Bradley \(1976\)](#) and [Cattelan \(2012\)](#).

Since the objects in a paired comparison experiment are typically characterized by several experimental variables, the problem of design arises naturally, specifically concerning how the settings of these variables should be chosen for the experiment. How this choice affects various aspects of the statistical efficiency of the analysis can be investigated before data are collected. To this end, optimum experimental design theory ([Kiefer 1959, 1974](#); [Pázman 1986](#); [Pukelsheim 1993](#)) provides a framework for developing experimental designs that use the information in the data effectively to achieve specific goals, such as, for example, maximizing the precision of parameter estimates.

In this paper, we consider optimal and efficient designs for linear paired comparisons, where the experimental variables are quantitative factors and the pairs used in the experiment lie in a constrained design region. These constraints are motivated by certain psychometric models for psychological paired comparison questionnaires where the quantitative factors correspond to distinct personality traits and the objects are statements, commonly referred to as items. Specifically, they mean that each item is associated with exactly one trait, and that pairs are formed only between items representing different traits (cf. [Brown and Maydeu-Olivares 2011](#), pp. 466–467).

We employ complete class results and invariance considerations to reduce the class of competing designs to a smaller set of designs for which the design weights depend on a single real parameter. By optimizing this parameter, we obtain optimal designs for several optimality criteria. Other designs from the reduced class which can be readily implemented in practice are shown to possess a high statistical efficiency. All proofs are presented in the appendix.

2 Preliminaries

Suppose the influence of K experimental variables on a continuous response is to be investigated in a paired comparison experiment. In accordance with common design terminology, we refer to the experimental variables and their values as factors and levels, respectively.

Let $Y(\mathbf{x})$ denote the continuous response for a pair $\mathbf{x} = (\mathbf{s}, \mathbf{t})$ of objects $\mathbf{s} = (s_1, \dots, s_K)$ and $\mathbf{t} = (t_1, \dots, t_K)$, characterized by K -tuples of factor levels in a set $\mathcal{T}$,

and let $\mathcal{X} \subseteq \mathcal{T} \times \mathcal{T}$ be the design region of all pairs that can be used in the experiment. The linear paired comparison model can then be concisely expressed by the model equation

$$Y(\mathbf{x}) = \mathbf{F}(\mathbf{x})^\top \boldsymbol{\beta} + \varepsilon(\mathbf{x}), \quad (1)$$

where $\mathbf{F} : \mathcal{X} \rightarrow \mathbb{R}^p$ is a vector of p real-valued functions, $\boldsymbol{\beta} \in \mathbb{R}^p$ denotes an unknown parameter vector and $\varepsilon(\mathbf{x})$ is a random error variable with expected value $E(\varepsilon(\mathbf{x})) = 0$ and variance $\text{Var}(\varepsilon(\mathbf{x})) = \sigma^2$ for all $\mathbf{x} \in \mathcal{X}$. We assume that for not necessarily distinct pairs $\mathbf{x}_i$ and $\mathbf{x}_j$ with $i \neq j$ the errors $\varepsilon(\mathbf{x}_i)$ and $\varepsilon(\mathbf{x}_j)$ are uncorrelated.

The model specification is completed by assuming that $\mathbf{F}(\mathbf{x}) = \mathbf{f}(\mathbf{s}) - \mathbf{f}(\mathbf{t})$ for every $\mathbf{x} = (\mathbf{s}, \mathbf{t}) \in \mathcal{X}$, where $\mathbf{f} = (f_1, \dots, f_p)^\top$ consists of known regression functions $f_i : \mathcal{T} \rightarrow \mathbb{R}$. Hence, the expected value $E(Y(\mathbf{x})) = \mathbf{F}(\mathbf{x})^\top \boldsymbol{\beta} = (\mathbf{f}(\mathbf{s}) - \mathbf{f}(\mathbf{t}))^\top \boldsymbol{\beta}$ is decomposed into the difference between $\mathbf{f}(\mathbf{s})^\top \boldsymbol{\beta}$ and $\mathbf{f}(\mathbf{t})^\top \boldsymbol{\beta}$ which are often interpreted as means in a linear model for the not directly observable utility of the objects.

The linear paired comparison model is a general linear model with standard assumptions for the errors. We are interested in deriving optimal designs for estimating the parameter vector $\boldsymbol{\beta}$ by means of the ordinary least squares estimator, which is the best linear unbiased estimator for $\boldsymbol{\beta}$ in the model. In order to ensure that this is a well-posed problem, we make the mild assumptions that the design region $\mathcal{X}$ is compact and that the regression functions $f_1, \dots, f_p$ are linearly independent and continuous on $\mathcal{X}$.

Generalizing the notion of exact designs ξ_N of size N , which are lists of N not necessarily distinct elements of the design region $\mathcal{X}$ at which observations should be taken, in optimum design theory, approximate designs ξ are considered. These are defined as finitely supported probability measures on $\mathcal{X}$. Therefore, every approximate design ξ can be specified by enumerating a finite number of distinct points $\mathbf{x}_1, \dots, \mathbf{x}_J \in \mathcal{X}$ together with nonnegative design weights $w_j = \xi(\{\mathbf{x}_j\})$ where $\sum_{j=1}^J w_j = 1$. The support of ξ is then the set $\text{supp}(\xi) = \{\mathbf{x}_1, \dots, \mathbf{x}_J\}$. Depending on the design, the number of support points J can vary. We denote the set of all approximate designs on $\mathcal{X}$ by Ξ .

Moreover, we need to equip the design space $\mathcal{X}$ with a suitable σ -field $\mathfrak{B}$. When $\mathcal{X}$ is finite, this is simply the power set $\mathfrak{B} = \mathfrak{P}(\mathcal{X})$. If $\mathcal{X}$ is a subset of an Euclidean space, we use $\mathfrak{B} = \mathfrak{B}(\mathcal{X})$ where $\mathfrak{B}(\mathcal{X})$ contains the Borel sets in $\mathcal{X}$. In both cases, for every $\xi \in \Xi$ and every $B \in \mathfrak{B}$ we have $\xi(B) = \sum_{\mathbf{x} \in \text{supp}(\xi) \cap B} \xi(\{\mathbf{x}\})$.

The quality of an approximate design ξ is reflected by its information matrix $\mathbf{M}(\xi)$. In the linear paired comparison model (1), for a design ξ with design points $\mathbf{x}_j = (\mathbf{s}_j, \mathbf{t}_j)$ and weights w_j for $j = 1, \dots, J$ this is given by

$$\mathbf{M}(\xi) = \int_{\mathcal{X}} \mathbf{F}\mathbf{F}^\top d\xi = \sum_{j=1}^J w_j \mathbf{F}(\mathbf{x}_j) \mathbf{F}(\mathbf{x}_j)^\top = \sum_{j=1}^J w_j (\mathbf{f}(\mathbf{s}_j) - \mathbf{f}(\mathbf{t}_j)) (\mathbf{f}(\mathbf{s}_j) - \mathbf{f}(\mathbf{t}_j))^\top.$$

Both the sets Ξ and $\mathcal{M}(\Xi) = \{\mathbf{M}(\xi) : \xi \in \Xi\}$ are convex. An important partial order on the set of information matrices is the Loewner ordering, which, for any symmetric matrices $\mathbf{A}$ and $\mathbf{B}$ of the same dimension, is defined by $\mathbf{A} \geq \mathbf{B}$ if and only if $\mathbf{A} - \mathbf{B}$ is positive semidefinite. In what follows, for brevity, the term ‘design’ is always understood to refer to an approximate design unless specified otherwise.

The information matrix $\mathbf{M}(\xi)$ of any design ξ is symmetric and positive semidefinite and hence possesses p not necessarily distinct real eigenvalues $\lambda_1(\xi), \dots, \lambda_p(\xi) \geq 0$ which can occur in any order. We are interested in deriving designs ξ that are optimal with regard to optimality criteria $\Phi : \Xi \rightarrow \mathbb{R}$ that depend on ξ only through the vector of eigenvalues $\boldsymbol{\lambda}(\xi) = (\lambda_1(\xi), \dots, \lambda_p(\xi))^\top$. Hence these criteria take the form $\Phi(\xi) = \varphi(\boldsymbol{\lambda}(\xi))$ for a suitable function φ and all $\xi \in \Xi$. We consider criteria Φ for which the optimum design problem can be stated as finding a design ξ^* such that $\Phi(\xi^*) \geq \Phi(\xi)$ holds for all $\xi \in \Xi$. A corresponding design ξ^* is called Φ -optimal. In general, a Φ -optimal design is not necessarily unique.

A broad class of criteria of the above type is obtained by choosing φ as a vector mean φ_q , where $-\infty \leq q \leq 1$ (Pukelsheim 1993, p. 139), resulting in the Φ_q -criterion with $\Phi_q(\xi) = \varphi_q(\boldsymbol{\lambda}(\xi))$ for all $\xi \in \Xi$. The Φ_q -criteria are isotonic with respect to the Loewner ordering on $\mathcal{M}(\Xi)$ in the sense that $\mathbf{M}(\xi) \geq \mathbf{M}(\xi')$ implies $\Phi_q(\xi) \geq \Phi_q(\xi')$. Moreover, they are concave on Ξ which follows from the properties of the vector means φ_q (cf. Pukelsheim 1993, p. 119).

The Φ_q -criteria coincide with Kiefer's simple trace criteria (Kiefer 1974, p. 862) and include as special cases the classical criteria of D -optimality (maximizing the determinant of $\mathbf{M}(\xi)$) for $q = 0$, A -optimality (minimizing for regular $\mathbf{M}(\xi)$ the trace of $\mathbf{M}(\xi)^{-1}$) for $q = -1$, and E -optimality (maximizing the smallest eigenvalue of $\mathbf{M}(\xi)$) for $q = -\infty$. For later use we note that $\Phi_0(\xi) = \varphi_0(\boldsymbol{\lambda}(\xi)) = (\prod_{i=1}^p \lambda_i(\xi))^{1/p}$ for D -optimality,

$$\Phi_{-1}(\xi) = \varphi_{-1}(\boldsymbol{\lambda}(\xi)) = \begin{cases} \left(\frac{1}{p} \sum_{i=1}^p \frac{1}{\lambda_i(\xi)} \right)^{-1} & \text{if } \mathbf{M}(\xi) \text{ is regular,} \\ 0 & \text{otherwise} \end{cases}$$

for A -optimality, and $\Phi_{-\infty}(\xi) = \varphi_{-\infty}(\boldsymbol{\lambda}(\xi)) = \min(\lambda_1(\xi), \dots, \lambda_p(\xi))$ for E -optimality. A set $\tilde{\Xi} \subset \Xi$ of designs is an essentially complete class for a Φ_q -criterion on Ξ , if for every $\eta \in \Xi$ there exists a $\xi \in \tilde{\Xi}$ with $\Phi_q(\xi) \geq \Phi_q(\eta)$.

Given a Φ_q -optimal design $\xi^* \in \Xi$, the performance of any other design $\xi \in \Xi$ can be assessed by means of the efficiency

$$\text{eff}_q(\xi) = \frac{\Phi_q(\xi)}{\Phi_q(\xi^*)}. \quad (2)$$

Since $0 \leq \text{eff}_q(\xi) \leq 1$, the efficiency is frequently interpreted as a percentage. Considering the efficiency $\text{eff}_q(\xi)$ often allows for the identification of a design $\xi \in \Xi$ that performs well, but which may be easier to implement in practice than a Φ_q -optimal design ξ^* .

For later use we note that any given exact design ξ_N of arbitrary size N can be identified with the approximate design $\tilde{\xi}_N \in \Xi$ whose support consists of the distinct points $\mathbf{x}_1, \dots, \mathbf{x}_J$ that are used by ξ_N , and for which $\tilde{\xi}_N(\{\mathbf{x}_j\}) = n_j/N$, where n_j is the number of replications of $\mathbf{x}_j$ in ξ_N . By using $\tilde{\xi}_N$, the exact design ξ_N can be regarded as an approximate design allowing its efficiency to be calculated as for any other design in Ξ .

We also remind the reader that the convex hull $\text{conv}(S)$ of a set $S \subseteq \mathbb{R}^n$ is the intersection of all convex sets that contain S as a subset. Hence $\text{conv}(S)$ is the smallest convex set containing S . An extreme point of a closed convex set $C \subseteq \mathbb{R}^n$ is any $\mathbf{w} \in C$ that cannot be expressed as a convex combination $\alpha \mathbf{u} + (1 - \alpha) \mathbf{v}$ with $0 < \alpha < 1$ of two distinct $\mathbf{u}, \mathbf{v} \in C$.

3 Designs for a constrained design region

When all explanatory variables are qualitative factors, the design problem for linear paired comparisons has been extensively studied by Graßhoff et al. (2003), Graßhoff et al. (2004), Großmann et al. (2006), and Großmann et al. (2009), among others. Moreover, results for the linear paired comparison model also apply to discrete choice experiments with a binary response when $\beta = \mathbf{0}$ is assumed in model (1) at the design stage (for an overview, see Großmann and Schwabe 2015).

By contrast, in what follows, we consider linear paired comparison experiments where all $K \geq 2$ factors are quantitative variables. Challenges for deriving optimal designs arise from a constraint on the pairs that can be used. Specifically, the experiments we consider are characterized by the condition that within each pair $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$ of objects $\mathbf{s} = (s_1, \dots, s_K)$ and $\mathbf{t} = (t_1, \dots, t_K)$ exactly one level s_k and exactly one level t_ℓ , where $k \neq \ell$, lies in a bounded interval $[a, b] \subseteq \mathbb{R}$ with $0 < a < b$ of positive real numbers, while all other levels are set to zero. In other words, a single but different factor is used to specify the positive level for both objects. Which factors $1 \leq k \neq \ell \leq K$ are used to this end, varies between pairs. Hence, denoting the i th canonical unit vector in $\mathbb{R}^K$ by $\mathbf{e}_i$, the pairs considered are $(\mathbf{s}, \mathbf{t})$ with $\mathbf{s} = s_k \mathbf{e}_k$ and $\mathbf{t} = t_\ell \mathbf{e}_\ell$ where $k \neq \ell$ and the design region is given by

$$\mathcal{X} = \bigcup_{k \neq \ell} \{(s_k \mathbf{e}_k, t_\ell \mathbf{e}_\ell) : a \leq s_k, t_\ell \leq b\}. \quad (3)$$

For every pair $(\mathbf{s}, \mathbf{t}) = (s_k \mathbf{e}_k, t_\ell \mathbf{e}_\ell) \in \mathcal{X}$, the influence of the factors on the response $Y(\mathbf{s}, \mathbf{t})$ is modelled by using $\mathbf{F}(\mathbf{s}, \mathbf{t}) = \mathbf{f}(\mathbf{s}) - \mathbf{f}(\mathbf{t}) = \mathbf{s} - \mathbf{t} = s_k \mathbf{e}_k - t_\ell \mathbf{e}_\ell$ in (1) so that the number of model parameters is $p = K$. The set $\mathbf{F}(\mathcal{X}) = \{\mathbf{F}(\mathbf{s}, \mathbf{t}) : (\mathbf{s}, \mathbf{t}) \in \mathcal{X}\}$ is known as the induced design region. Since for any $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$ we have $(\mathbf{t}, \mathbf{s}) \in \mathcal{X}$ and $\mathbf{F}(\mathbf{t}, \mathbf{s}) = -\mathbf{F}(\mathbf{s}, \mathbf{t})$, it follows that $-\mathbf{F}(\mathcal{X}) = \{-\mathbf{F}(\mathbf{s}, \mathbf{t}) : (\mathbf{s}, \mathbf{t}) \in \mathcal{X}\}$ coincides with $\mathbf{F}(\mathcal{X})$. Figure 1 shows an example of the induced design region for $K = 3$ factors.

Our strategy for deriving optimal designs for the linear paired comparison model (1) with design region $\mathcal{X}$ in (3) relies on two main ideas. First, the set of designs that need to be considered is reduced from the set Ξ of all designs on $\mathcal{X}$ to a smaller class of designs. For every Φ_q -criterion with $q \in [-\infty, 1]$, this class is essentially complete. In particular, the class is guaranteed to contain a design that is Φ_q -optimal among all designs in Ξ . Secondly, optimal designs for selected Φ_q -criteria are identified within the smaller class.

The basis for a first reduction of the set of competing designs Ξ is a result on matrix majorization originally proved by Ehrenfeld (1956) in the context of a specific optimality criterion. This result has been restated in general terms by Pukelsheim (1993, Theorem 8.5) and can be combined with the isotonicity of the Φ_q -criteria with

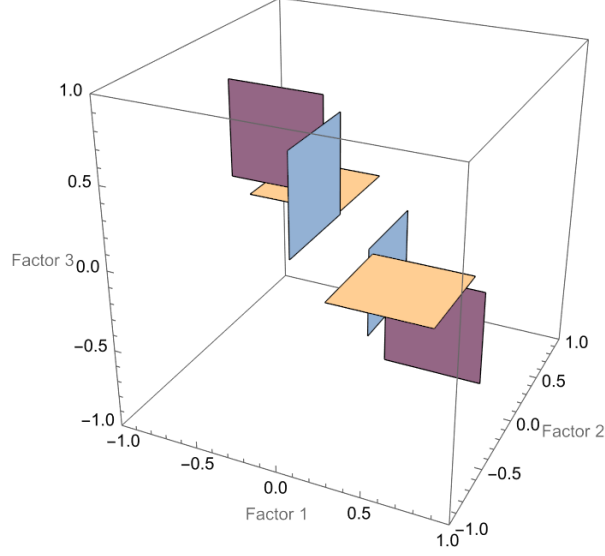


Fig. 1 Induced design region $\mathbf{F}(\mathcal{X})$ for $K = 3$ and $[a, b] = [0.2, 0.8]$

respect to the Loewner ordering on the set $\mathcal{M}(\Xi)$ of information matrices to give an essentially complete class of designs for all Φ_q -criteria with $q \in [-\infty, 1]$. The supports of the corresponding designs consist of points $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$ for which the vectors $\mathbf{F}(\mathbf{s}, \mathbf{t})$ are extreme points of the convex hull $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}) \cup -\mathbf{F}(\mathcal{X}))$ of the union of the induced design region $\mathbf{F}(\mathcal{X})$ and its reflection $-\mathbf{F}(\mathcal{X})$ through the origin. In other words, for every pair $(\mathbf{s}, \mathbf{t})$ in the support of such a Φ_q -optimal design there do not exist distinct vectors $\mathbf{u}, \mathbf{v} \in \mathcal{E}$ such that $\mathbf{F}(\mathbf{s}, \mathbf{t}) = \alpha \mathbf{u} + (1 - \alpha) \mathbf{v}$ for some $0 < \alpha < 1$. The set $\mathcal{E}$ is known as the Elfving set. For the present design problem where $\mathbf{F}(\mathcal{X}) = -\mathbf{F}(\mathcal{X})$ this simplifies to $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}))$.

In order to characterize the support points of the resulting designs we introduce the two disjoint subsets

$$\mathcal{O}_{a,b} = \{(s_k \mathbf{e}_k, t_\ell \mathbf{e}_\ell) \in \mathcal{X} : \{s_k, t_\ell\} = \{a, b\}, 1 \leq k \neq \ell \leq K\}$$

and

$$\mathcal{O}_b = \{(b \mathbf{e}_k, b \mathbf{e}_\ell) \in \mathcal{X} : 1 \leq k \neq \ell \leq K\}$$

of the design region $\mathcal{X}$ given by (3). The sizes of the two sets are $|\mathcal{O}_{a,b}| = 2K(K-1)$ and $|\mathcal{O}_b| = K(K-1)$, respectively. We refer to elements of $\mathcal{O}_{a,b}$ as pairs of type (a, b) and similarly to elements of $\mathcal{O}_b$ as pairs of type (b, b) . Moreover, we let $\mathcal{X}' = \mathcal{O}_{a,b} \cup \mathcal{O}_b$ denote the set of all pairs of type (a, b) or (b, b) and

$$\Xi' = \{\xi \in \Xi : \text{supp}(\xi) \subseteq \mathcal{X}'\} \quad (4)$$

denote the class of designs ξ whose support consists of pairs of type (a, b) , type (b, b) , or both.

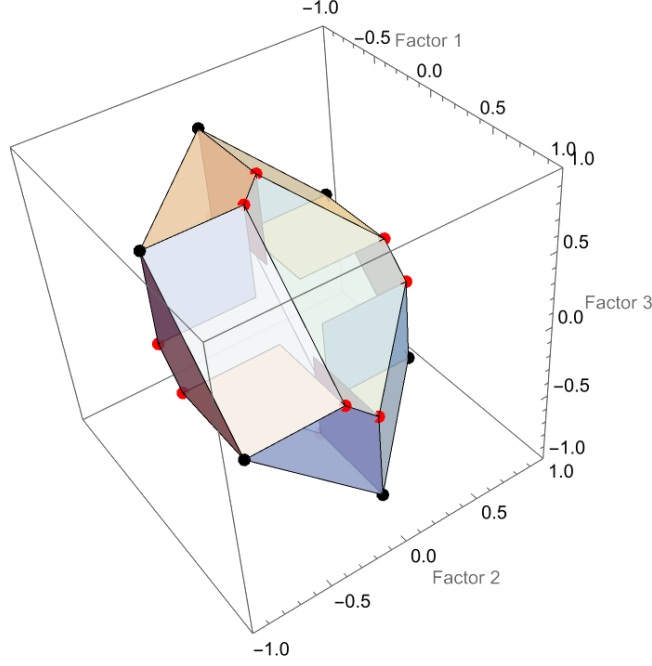


Fig. 2 Induced design region $\mathbf{F}(\mathcal{X})$ and Elfving set $\mathcal{E}$ for $K = 3$ and $[a, b] = [0.2, 0.8]$. Dots depict $\mathbf{F}(\mathbf{s}, \mathbf{t})$ for $(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_{a,b}$ (red) or $(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_b$ (black)

The following and all subsequent results refer to the linear paired comparison model (1) with design region $\mathcal{X}$ given by (3) and $\mathbf{F}(\mathbf{s}, \mathbf{t}) = \mathbf{s} - \mathbf{t}$ for all $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$.

The convex hull $\text{conv}(\mathbf{F}(\mathcal{X}'))$ of the set $\mathbf{F}(\mathcal{X}') = \{\mathbf{F}(\mathbf{s}, \mathbf{t}) : (\mathbf{s}, \mathbf{t}) \in \mathcal{X}'\}$ of regression vectors for points of type (a, b) or (b, b) is obviously a subset of the Elfving set $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}))$. The following lemma shows that the two sets are in fact equal and that the extreme points of $\mathcal{E}$ are contained in $\mathbf{F}(\mathcal{X}')$.

Lemma 1 *Let $\mathbf{F}(\mathcal{X}')$ be defined as above and $\tilde{\mathcal{E}}$ be the set of extreme points of the Elfving set $\mathcal{E}$. Then $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}'))$ and $\tilde{\mathcal{E}} \subseteq \mathbf{F}(\mathcal{X}')$.*

Figure 2 shows an example of the Elfving set $\mathcal{E}$ for $K = 3$ factors. The vertices of $\mathcal{E}$ represent the elements of the set $\mathbf{F}(\mathcal{X}')$.

With these preparations, we obtain the result that the search for Φ_q -optimal designs can be confined to designs with support contained in the set $\mathcal{X}'$ of pairs of type (a, b) or type (b, b) .

Theorem 1 *The set Ξ' is an essentially complete class for every Φ_q -criterion on Ξ with $q \in [-\infty, 1]$.*

Theorem 1 ensures that a Φ_q -optimal design on $\mathcal{X}$ can be found within the class Ξ' of designs whose support consists of pairs $(s_k \mathbf{e}_k, t_\ell \mathbf{e}_\ell) \in \mathcal{X}' = \mathcal{O}_{a,b} \cup \mathcal{O}_b$ with $k \neq \ell$ for which the nonzero components are equal to $s_k = a$ and $t_\ell = b$, or $s_k = b$ and $t_\ell = a$, or $s_k = t_\ell = b$. Hence, for the class Ξ' , the infinite number of possible supports of optimal designs in Ξ is reduced to a finite number, since $|\mathcal{X}'| = |\mathcal{O}_{a,b}| + |\mathcal{O}_b| = 2K(K-1) + K(K-1) = 3K(K-1)$.

A further reduction of Ξ' to a smaller set Ξ^* of designs that is guaranteed to contain a Φ_q -optimal design can be achieved by using invariance considerations (see, e.g., Schwabe (1996, Chapter 3.2) or Pukelsheim (1993, Chapter 13)). To this end, in the proof of the following theorem, we consider a finite group $\mathcal{G}$ of one-to-one transformations of the design space $\mathcal{X}$ onto itself whose elements correspond to simultaneous permutations of the K factors in all pairs $(\mathbf{s}, \mathbf{t})$ or swapped pairs $(\mathbf{t}, \mathbf{s})$ in $\mathcal{X}$. The idea of the proof is then to consider for every design $\eta \in \Xi$ a design $\xi \in \Xi'$ from Theorem 1 with $\Phi_q(\xi) \geq \Phi_q(\eta)$ and to show that the so-called symmetrization $\bar{\xi}$ of ξ with respect to $\mathcal{G}$ (Schwabe 1996, p. 21) is an element of Ξ^* with $\Phi_q(\bar{\xi}) \geq \Phi_q(\xi)$.

The pertinent class of designs is

$$\Xi^* = \{\xi_w \in \Xi' : w \in [0, 1]\},$$

where, for every $w \in [0, 1]$, the design $\xi_w \in \Xi'$ assigns weight $\xi_w(\{(\mathbf{s}, \mathbf{t})\}) = \frac{w}{|\mathcal{O}_{a,b}|}$ to every pair $(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_{a,b}$ and weight $\xi_w(\{(\mathbf{s}, \mathbf{t})\}) = \frac{1-w}{|\mathcal{O}_b|}$ to every pair $(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_b$.

Theorem 2 *The set $\Xi^* \subset \Xi'$ is an essentially complete class for every Φ_q -criterion on Ξ with $q \in [-\infty, 1]$.*

Theorem 2 shows that for finding Φ_q -optimal designs on $\mathcal{X}$ only the maximization of $\Phi_q(\xi_w)$ for designs $\xi_w \in \Xi^*$ needs to be considered. The value of $\Phi_q(\xi_w)$ solely depends on the eigenvalues $\lambda(\xi_w)$ of the information matrix $\mathbf{M}(\xi_w)$ which can be readily determined due to the simple structure of ξ_w . In what follows, $\mathbf{I}_K$ and $\mathbf{J}_K$ denote the identity matrix and the all-one matrix of order $K \times K$.

Lemma 2 *Let $c = a/b$ and $\alpha(c, K) = (1-c)^2 + (1-c^2)K$. The information matrix of any design $\xi_w \in \Xi^*$ is the completely symmetric matrix*

$$\mathbf{M}(\xi_w) = \frac{b^2}{K(K-1)} (((K-1)m_1(w) + 2m_2(w)) \mathbf{I}_K - 2m_2(w) \mathbf{J}_K), \quad (5)$$

where $m_1(w) = 2 - (1-c^2)w$ and $m_2(w) = 1 - (1-c)w$. The eigenvalues of $\mathbf{M}(\xi_w)$ are

$$\lambda_1(\xi_w) = \frac{b^2}{K} (m_1(w) - 2m_2(w)) = \frac{b^2}{K} (1-c)^2 w \quad (6)$$

with multiplicity 1 and

$$\lambda_2(\xi_w) = \frac{b^2}{K(K-1)} ((K-1)m_1(w) + 2m_2(w)) = \frac{b^2}{K(K-1)} (2K - \alpha(c, K)w) \quad (7)$$

with multiplicity $K-1$. Moreover, $0 \leq \lambda_1(\xi_w) < \lambda_2(\xi_w)$.

As an obvious consequence of Lemma 2, the information matrix $\mathbf{M}(\xi_0)$ of the design $\xi_0 \in \Xi^*$ that uses only pairs of type (b, b) is singular. Therefore, the vector β of model parameters in (1) is not linearly estimable when $\xi_0 \in \Xi^*$ is implemented in practice.

The designs ξ_w in the essentially complete class Ξ^* are uniquely determined by the mass $w = \xi_w(\mathcal{O}_{a,b})$ they assign to the set of all pairs of type (a, b) . Consequently, in order to obtain an optimal design ξ_w for a specific Φ_q -criterion, only the value of w needs to be optimized. Lemma 2 shows that a corresponding Φ_q -optimal design does not depend on the specific range $[a, b]$ used for the positive factor levels but only on the ratio $c = a/b$.

By using this approach, for the Φ_q -criteria corresponding to A -, D -, and E -optimality we can derive optimal designs in Ξ^* . In view of the essential completeness of Ξ^* , these designs are however also Φ_q -optimal among all designs in Ξ .

Theorem 3 *Let $c = a/b$ and $\alpha(c, K) = (1 - c)^2 + (1 - c^2)K$. For $q \in \{-\infty, -1, 0\}$ the design $\xi_{w_q} \in \Xi^*$, where*

$$w_{-\infty} = 1, \\ w_{-1} = \min \left(\frac{2K}{\alpha(c, K) + (1 - c)(K - 1)\sqrt{\alpha(c, K)}}, 1 \right)$$

and

$$w_0 = \min \left(\frac{2}{\alpha(c, K)}, 1 \right)$$

is Φ_q -optimal in Ξ .

The E -optimal design $\xi_{w_{-\infty}}$ in Theorem 3 uses exclusively pairs of type (a, b) , while for the A - and D -optimal designs $\xi_{w_{-1}}$ and ξ_{w_0} the total masses w_{-1} and w_0 on pairs of this type depend on the ratio $c = a/b$ of the smallest to the largest positive factor level and the number of factors K . Figures 3 and 4 illustrate this dependence.

For both criteria, it can be seen that for every K the optimal designs assign all mass to the pairs of type (a, b) if c is equal to or exceeds a certain threshold. For the A -criterion this threshold $c_{-1}(K)$ is a root of a quartic equation in c and for the D -criterion it can be calculated as $c_0(K) = (\sqrt{1 + (K - 1)^2} - 1)/(K - 1)$. In principle, $c_{-1}(K)$ can be expressed in closed form, but the derivations are cumbersome.

The figures also show that for every fixed $0 < c < 1$, the weights w_{-1} and w_0 decrease if K gets larger. Overall, for $K \leq 10$ and most values of c , the A -optimal design $\xi_{w_{-1}}$ tends to use a considerably larger proportion of pairs of type (a, b) than of type (b, b) . For the D -optimal design ξ_{w_0} this tendency is reversed, if K is moderately large.

As noted, for $c \geq c_{-1}(K)$ the A -optimal design $\xi_{w_{-1}}$ coincides with the uniform design $\xi_1 \in \Xi^*$ that splits the total weight equally between the pairs of type (a, b) . Similarly, we have $\xi_{w_0} = \xi_1$ for the D -optimal design ξ_{w_0} if $c \geq c_0(K)$. However, for values of c that are smaller than the thresholds $c_{-1}(K)$ or $c_0(K)$, using only pairs of type (a, b) entails some efficiency loss. In particular, considering the efficiency (2)

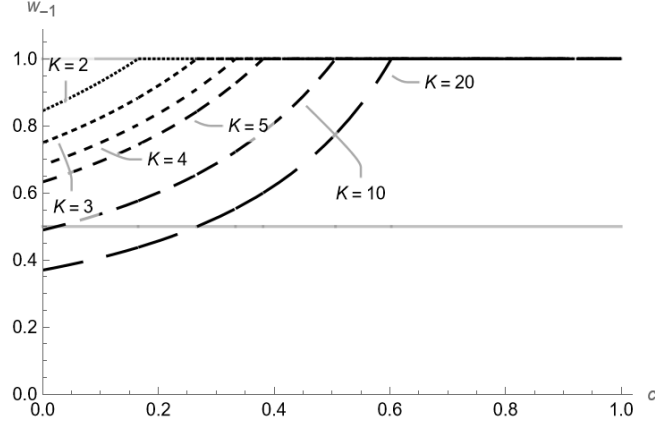


Fig. 3 Weights w_{-1} of A -optimal designs vs. $c = a/b$ for various K with reference line at 0.5

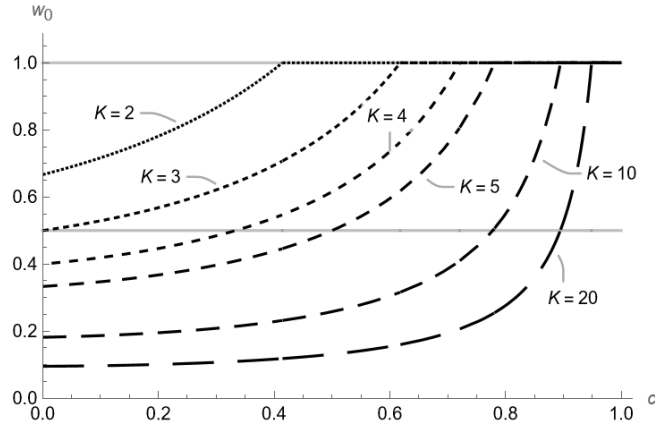


Fig. 4 Weights w_0 of D -optimal designs vs. $c = a/b$ for various K with reference line at 0.5

with $\xi^* = \xi_{w_{-1}}$ or $\xi^* = \xi_{w_0}$ in the denominator respectively, we have $\text{eff}_{-1}(\xi_1) < 1$ for $c < c_{-1}(K)$ and $\text{eff}_0(\xi_1) < 1$ for $c < c_0(K)$.

If c is close to 0, then $w_{-1} \approx 2K/(K+1+(K-1)\sqrt{K+1})$ and $w_0 \approx 2/(K+1)$. The corresponding designs $\xi_{2K/(K+1+(K-1)\sqrt{K+1})} \in \Xi^*$ and $\xi_{2/(K+1)} \in \Xi^*$ may then respectively perform better in terms of the A - or D -optimality than ξ_1 for values of c where the latter is not optimal. Figure 5 depicts the efficiencies $\text{eff}_{-1}(\xi_{2K/(K+1+(K-1)\sqrt{K+1})})$ and $\text{eff}_{-1}(\xi_1)$ for the A -criterion. Likewise, Figure 6 shows the efficiencies $\text{eff}_0(\xi_{2/(K+1)})$ and $\text{eff}_0(\xi_1)$ for the D -criterion. Moreover, the figures also present the efficiency of the design $\xi_{1/2} \in \Xi^*$ for comparison, as this uses the pairs of types (a, b) and (b, b) in equal proportions.

Figures 5 and 6 illustrate that for both criteria the designs ξ_1 and the other designs complement each other in the sense that for values of c for which the efficiency of ξ_1 is

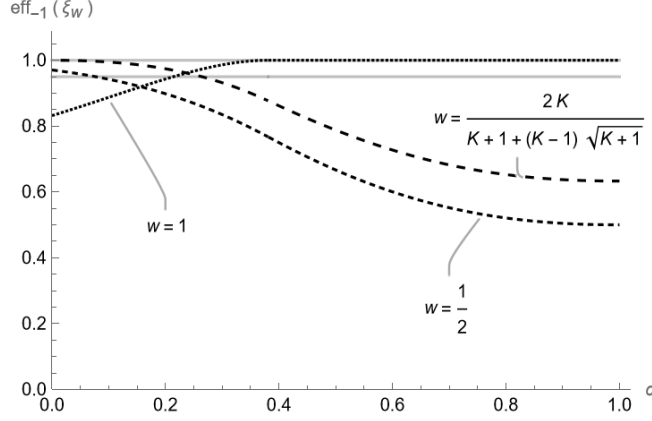


Fig. 5 Efficiency $\text{eff}_{-1}(\xi_w)$ for A -criterion vs. $c = a/b$ for different $\xi_w \in \Xi^*$ and $K = 5$ with reference line at 0.95

relatively low, the efficiency of at least one of the other designs is high. In particular, for every $0 < c < 1$ the efficiency of at least one of ξ_1 , $\xi_{1/2}$, $\xi_{2K/(K+1+(K-1)\sqrt{K+1})}$, or $\xi_{2/(K+1)}$ is greater than or equal to 0.95. Although in both figures we consider $K = 5$ factors, other values of K produce results that are qualitatively similar. However, for some values of c , the efficiencies of all the considered designs can be smaller than 0.95. Which of the designs achieves the highest efficiency for a given value of c depends on K and the criterion.

The design ξ_1 can be readily implemented as an exact design in practice by equally replicating the $2K(K-1)$ pairs of type (a, b) . Likewise, the design $\xi_{1/2}$ can be implemented by replicating each pair of type (a, b) once and each of the $K(K-1)$ pairs of type (b, b) twice. Similar implementations are possible for the design $\xi_{2/(K+1)}$ and, after approximating the total weight assigned to the pairs of type (a, b) with a rational number, for $\xi_{2K/(K+1+(K-1)\sqrt{K+1})}$.

Moreover, there exist efficient exact designs that use only pairs from a proper subset of $\mathcal{X}' = \mathcal{O}_{a,b} \cup \mathcal{O}_b$. When regarded as approximate designs, these possess the same information matrices, and hence the same A - and D -efficiencies, as the designs ξ_1 , $\xi_{1/2}$, and $\xi_{2/(K+1)}$. For the design $\xi_{2K/(K+1+(K-1)\sqrt{K+1})}$, a similar statement holds only approximately.

To illustrate, consider the case where the number of factors K is odd. For $k = 1, \dots, K$ and $\ell = 1, \dots, (K-1)/2$ let

$$m(k, \ell) = \begin{cases} k + \ell & \text{if } k + \ell \leq K, \\ k + \ell - K & \text{if } k + \ell > K. \end{cases}$$

For any design $\xi_w \in \Xi^*$ with rational $0 < w \leq 1$, an exact design of size N whose efficiency matches that of ξ_w can then be obtained by replicating all $K(K-1)/2$ points

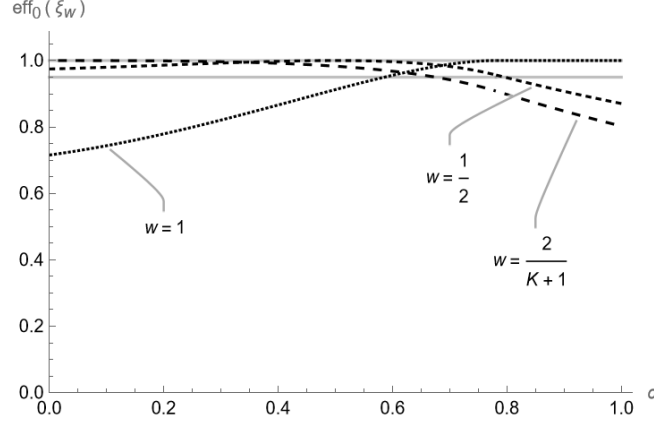


Fig. 6 Efficiency $\text{eff}_0(\xi_w)$ for D -criterion vs. $c = a/b$ for different $\xi_w \in \Xi^*$ and $K = 5$ with reference line at 0.95

Table 1 Exact design of size $N = 30$ for $K = 5$ and $[a, b] = [0.2, 0.8]$ with same D -efficiency as $\xi_{2/(K+1)}$

Pair	Settings	Pair	Settings
1	((0.8, 0, 0, 0, 0), (0, 0.2, 0, 0, 0))	16	((0.8, 0, 0, 0, 0), (0, 0, 0.8, 0, 0))
2	((0, 0.8, 0, 0, 0), (0, 0, 0.2, 0, 0))	17	((0, 0.8, 0, 0, 0), (0, 0, 0, 0.8, 0))
3	((0, 0, 0.8, 0, 0), (0, 0, 0, 0.2, 0))	18	((0, 0, 0.8, 0, 0), (0, 0, 0, 0, 0.8))
4	((0, 0, 0, 0.8, 0), (0, 0, 0, 0, 0.2))	19	((0, 0, 0, 0.8, 0), (0.8, 0, 0, 0, 0))
5	((0, 0, 0, 0, 0.8), (0.2, 0, 0, 0, 0))	20	((0, 0, 0, 0, 0.8), (0, 0.8, 0, 0, 0))
6	((0.8, 0, 0, 0, 0), (0, 0, 0.2, 0, 0))	21	((0.8, 0, 0, 0, 0), (0, 0.8, 0, 0, 0))
7	((0, 0.8, 0, 0, 0), (0, 0, 0, 0.2, 0))	22	((0, 0.8, 0, 0, 0), (0, 0, 0.8, 0, 0))
8	((0, 0, 0.8, 0, 0), (0, 0, 0, 0, 0.2))	23	((0, 0, 0.8, 0, 0), (0, 0, 0, 0.8, 0))
9	((0, 0, 0, 0.8, 0), (0.2, 0, 0, 0, 0))	24	((0, 0, 0, 0.8, 0), (0, 0, 0, 0, 0.8))
10	((0, 0, 0, 0, 0.8), (0, 0.2, 0, 0, 0))	25	((0, 0, 0, 0, 0.8), (0.8, 0, 0, 0, 0))
11	((0.8, 0, 0, 0, 0), (0, 0.8, 0, 0, 0))	26	((0.8, 0, 0, 0, 0), (0, 0, 0.8, 0, 0))
12	((0, 0.8, 0, 0, 0), (0, 0, 0.8, 0, 0))	27	((0, 0.8, 0, 0, 0), (0, 0, 0, 0.8, 0))
13	((0, 0, 0.8, 0, 0), (0, 0, 0, 0.8, 0))	28	((0, 0, 0.8, 0, 0), (0, 0, 0, 0, 0.8))
14	((0, 0, 0, 0.8, 0), (0, 0, 0, 0, 0.8))	29	((0, 0, 0, 0.8, 0), (0.8, 0, 0, 0, 0))
15	((0, 0, 0, 0, 0.8), (0.8, 0, 0, 0, 0))	30	((0, 0, 0, 0, 0.8), (0, 0.8, 0, 0, 0))

in each of

$$\mathcal{O}_{a,b}^{\subset} = \{(a\mathbf{e}_k, b\mathbf{e}_{m(k,\ell)}) : k = 1, \dots, K, \ell = 1, \dots, (K-1)/2\} \subset \mathcal{O}_{a,b}$$

and

$$\mathcal{O}_b^{\subset} = \{(b\mathbf{e}_k, b\mathbf{e}_{m(k,\ell)}) : k = 1, \dots, K, \ell = 1, \dots, (K-1)/2\} \subset \mathcal{O}_b$$

in proportions w and $1 - w$ of the total size, where N is chosen such that both N and Nw are divisible by $K(K-1)/2$. For example, for $K = 5$ and $[a, b] = [0.2, 0.8]$, an exact design corresponding to ξ_w with $w = 2/(K+1) = 1/3$ is shown in Table 1. This design of size $N = 30$ uses every pair in $\mathcal{O}_{a,b}^{\subset}$ once and every pair in $\mathcal{O}_b^{\subset}$ twice. Since $c = a/b = 0.25$, it can be seen from Figure 6 that the design is highly D -efficient.

4 Discussion

The optimum design problem we have considered is challenging due to the shape of the constrained design region. Nevertheless, by using a majorization argument and invariance considerations it is possible to narrow the search for optimal designs down from the set of all designs to a smaller class that is known to contain an optimum solution. This was achieved in two steps by establishing the complete class results in Theorems 1 and 2. These results are general and apply to all Φ_q -criteria with $q \in [-\infty, 1]$.

The designs in the resulting class Ξ^* have a simple structure and are concentrated on the pairs of types (a, b) or (b, b) . As a consequence, the eigenvalues of the information matrix on which the Φ_q -criteria are based can be easily determined, as shown by Lemma 2. Moreover, the designs in Ξ^* are uniquely determined by the total weight they place on the pairs of type (a, b) . This simplifies the optimization problem considerably. For the Φ_q -criteria corresponding to the classical criteria of E -, A - and D -optimality we derived explicit solutions in Theorem 3. For other Φ_q -criteria with $q \in [-\infty, 1]$, optimal designs can be obtained along the same lines, although we have not pursued this here.

The E -optimal design in Theorem 3 uses exclusively pairs of type (a, b) , while the A - and D -optimal designs in the theorem generally employ a mixture of pairs of both types (a, b) and (b, b) . However, depending on the ratio $c = a/b$ of the values $0 < a < b$ that constrain the positive factor levels and on the number of factors K the two latter designs frequently also put all mass on the pairs of type (a, b) , as illustrated in Figures 3 and 4.

This observation motivated efficiency comparisons for the A - and D -criterion involving the uniform design ξ_1 on the pairs of type (a, b) , the design $\xi_{1/2}$ which uses pairs of types (a, b) and (b, b) in equal proportions, and, for each of the criteria, another design. From the comparisons we conclude that there exist highly efficient designs that can be readily implemented in practice as exact designs. This is particularly useful since, typically, the weight w_{-1} that determines the A -optimal design in Theorem 3 is not a rational number. Despite this practical limitation, knowing the A -optimal design and optimal approximate designs in general is helpful, as these provide a benchmark for assessing the efficiency of alternative designs.

For the design problem and criteria considered, in conjunction with Theorem 1, Carathéodory's theorem (see, e.g., Silvey 1980, p. 72) guarantees the existence of optimal designs with at most $K(K+1)/2$ support points of type (a, b) or (b, b) . For $K \geq 2$ this number is smaller than the total number $3K(K-1)$ of pairs of the two types. Although by using the results in the current paper, in principle an algorithm for constructing E -, A -, or D -optimal designs with support of size $K(K+1)/2$ could be devised from the proof of Carathéodory's theorem, we have not pursued this idea. However, for $K = 3, 4$, and 5 we have been able to analytically construct A - and D -optimal designs with at most $K(K+1)/2$ support points where the specific number of points also depends on the value of $0 < c < 1$. Furthermore, for the same values of K and c we obtained equivalent D -optimal designs by using the algorithm of Harman et al. (2024), while for the A -criterion the algorithm is not directly applicable, because of the generally irrational weight w_{-1} .

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Appendix

In the appendix we present the proofs of the results in the body of the text.

Proof of Lemma 1 With the notation in the body of the paper, the Elfving set is $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}))$. For all $1 \leq k < \ell \leq K$ the set $\{\pm \mathbf{F}(s_k \mathbf{e}_k, t_\ell \mathbf{e}_\ell) : a \leq s_k, t_\ell \leq b\}$ is the union of a square in a two-dimensional plane of $\mathbb{R}^K$ with its reflection through the origin, as is illustrated in Figure 1. This implies $\text{conv}(\{\pm \mathbf{F}(s_k \mathbf{e}_k, t_\ell \mathbf{e}_\ell) : a \leq s_k, t_\ell \leq b\}) = \text{conv}(\{\pm \mathbf{F}(a \mathbf{e}_k, b \mathbf{e}_\ell), \pm \mathbf{F}(b \mathbf{e}_k, a \mathbf{e}_\ell), \pm \mathbf{F}(b \mathbf{e}_k, b \mathbf{e}_\ell)\})$, which is a hexagon in the same plane.

The equality $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}'))$ then follows from

$$\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X})) = \text{conv} \left(\bigcup_{k < \ell} \text{conv}(\{\pm \mathbf{F}(s_k \mathbf{e}_k, t_\ell \mathbf{e}_\ell) : a \leq s_k, t_\ell \leq b\}) \right)$$

and

$$\text{conv}(\mathbf{F}(\mathcal{X}')) = \text{conv} \left(\bigcup_{k < \ell} \text{conv}(\{\pm \mathbf{F}(a \mathbf{e}_k, b \mathbf{e}_\ell), \pm \mathbf{F}(b \mathbf{e}_k, a \mathbf{e}_\ell), \pm \mathbf{F}(b \mathbf{e}_k, b \mathbf{e}_\ell)\}) \right).$$

In view of $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}'))$, the set $\tilde{\mathcal{E}}$ of extreme points of $\mathcal{E}$ coincides with the set of extreme points of $\text{conv}(\mathbf{F}(\mathcal{X}'))$. Because $\mathbf{F}(\mathcal{X}')$ is finite, $\text{conv}(\mathbf{F}(\mathcal{X}'))$ is a polytope. Therefore, the extreme points of $\text{conv}(\mathbf{F}(\mathcal{X}'))$ are contained in $\mathbf{F}(\mathcal{X}')$ (Brøndsted 1983, Theorem 7.2). This implies $\tilde{\mathcal{E}} \subseteq \mathbf{F}(\mathcal{X}')$. $\square$

Proof of Theorem 1 Let $q \in [-\infty, 1]$. We show that for every design $\eta \in \Xi$ there exists a design $\xi \in \Xi'$ with $\Phi_q(\xi) \geq \Phi_q(\eta)$.

The mapping $\mathbf{F} : \mathcal{X} \rightarrow \mathbf{F}(\mathcal{X})$ is bijective. This implies a corresponding bijection between the designs in Ξ and the designs on $\mathbf{F}(\mathcal{X})$. In particular, every design on $\mathbf{F}(\mathcal{X})$ is equal to the induced design $\zeta^{\mathbf{F}}$ with support points $\mathbf{F}(\mathbf{s}_1, \mathbf{t}_1), \dots, \mathbf{F}(\mathbf{s}_J, \mathbf{t}_J)$ and corresponding design weights $\zeta^{\mathbf{F}}(\{\mathbf{F}(\mathbf{s}_j, \mathbf{t}_j)\}) = w_j$ for a uniquely determined design $\zeta \in \Xi$ with support points $(\mathbf{s}_1, \mathbf{t}_1), \dots, (\mathbf{s}_J, \mathbf{t}_J)$ and design weights $\zeta(\{(\mathbf{s}_j, \mathbf{t}_j)\}) = w_j$ for $j = 1, \dots, J$.

As before, let $\tilde{\mathcal{E}}$ be the set of extreme points of the Elfving set $\mathcal{E} = \text{conv}(\mathbf{F}(\mathcal{X}))$. Consider a design $\eta \in \Xi$ and the induced design $\tilde{\eta} = \eta^{\mathbf{F}}$ on $\mathbf{F}(\mathcal{X})$. If the support of $\tilde{\eta}$ is contained in $\tilde{\mathcal{E}}$, there is nothing to show since $\tilde{\mathcal{E}} \subseteq \mathbf{F}(\mathcal{X}')$ by Lemma 1 which implies $\text{supp}(\eta) \subseteq \mathbf{F}^{-1}(\tilde{\mathcal{E}}) \subseteq \mathcal{X}'$ and $\eta \in \Xi'$.

If the support of $\tilde{\eta} = \eta^{\mathbf{F}}$ is not included in $\tilde{\mathcal{E}}$, then there exists a design $\tilde{\xi} = \xi^{\mathbf{F}}$ on $\mathbf{F}(\mathcal{X})$ whose support is included in $\tilde{\mathcal{E}}$ and for which $\int_{\mathbf{F}(\mathcal{X})} \tilde{\mathbf{x}} \tilde{\mathbf{x}}^\top d\tilde{\xi}(\tilde{\mathbf{x}}) \geq \int_{\mathbf{F}(\mathcal{X})} \tilde{\mathbf{x}} \tilde{\mathbf{x}}^\top d\tilde{\eta}(\tilde{\mathbf{x}})$ (Pukelsheim 1993, Theorem 8.5). It follows that

$$\mathbf{M}(\xi) = \int_{\mathcal{X}} \mathbf{F} \mathbf{F}^\top d\xi = \int_{\mathbf{F}(\mathcal{X})} \tilde{\mathbf{x}} \tilde{\mathbf{x}}^\top d\tilde{\xi}(\tilde{\mathbf{x}}) \geq \int_{\mathbf{F}(\mathcal{X})} \tilde{\mathbf{x}} \tilde{\mathbf{x}}^\top d\tilde{\eta}(\tilde{\mathbf{x}}) = \int_{\mathcal{X}} \mathbf{F} \mathbf{F}^\top d\eta = \mathbf{M}(\eta).$$

This implies $\Phi_q(\xi) \geq \Phi_q(\eta)$ due to the isotonicity of the Φ_q -criterion with respect to the Loewner ordering. Using Lemma 1 as before, yields $\text{supp}(\xi) \subseteq \mathcal{X}'$ and $\xi \in \Xi'$, which completes the proof. $\square$

Proof of Theorem 2 Let $q \in [-\infty, 1]$. We show that for every design $\eta \in \Xi$ there exists a design $\bar{\xi} \in \Xi^*$ with $\Phi_q(\bar{\xi}) \geq \Phi_q(\eta)$.

Let $\pi \in S_K$ be an element of the symmetric group of permutations of the numbers $1, \dots, K$ and define for every pair $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$ with $\mathbf{s} = (s_1, \dots, s_K)$ and $\mathbf{t} = (t_1, \dots, t_K)$ the mapping $g_\pi : \mathcal{X} \rightarrow \mathcal{X}$ by

$$g_\pi((s_1, \dots, s_K), (t_1, \dots, t_K)) = ((s_{\pi(1)}, \dots, s_{\pi(K)}), (t_{\pi(1)}, \dots, t_{\pi(K)})).$$

Moreover, define $\tilde{g}_\pi : \mathcal{X} \rightarrow \mathcal{X}$ by $\tilde{g}_\pi(\mathbf{s}, \mathbf{t}) = g_\pi(\mathbf{t}, \mathbf{s})$ for every $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$. The set $\mathcal{G} = \{g_\pi, \tilde{g}_\pi : \pi \in S_K\}$ together with the composition $\circ$ of mappings is a finite group of one-to-one transformations of $\mathcal{X}$ onto itself. Every element $g \in \mathcal{G}$ represents a simultaneous permutation of the K factors in all pairs $(\mathbf{s}, \mathbf{t})$ or swapped pairs $(\mathbf{t}, \mathbf{s})$ in $\mathcal{X}$.

For every $\xi \in \Xi$, every $g \in \mathcal{G}$ induces the design $\xi^g \in \Xi$ with $\xi^g(B) = \xi(g^{-1}(B))$ for every Borel set $B \subseteq \mathcal{X}$. If $\xi^g = \xi$ for every $g \in \mathcal{G}$, the design ξ is called invariant with respect to $\mathcal{G}$.

In addition to inducing designs, the group $\mathcal{G}$ induces linear transformations of the vector $\mathbf{F}$. That is, for every $g \in \mathcal{G}$ there exists a matrix $\mathbf{Q}_g$ such that $\mathbf{F}(g(\mathbf{s}, \mathbf{t})) = \mathbf{Q}_g \mathbf{F}(\mathbf{s}, \mathbf{t})$ for all $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$. For $g_\pi \in \mathcal{G}$ the matrix $\mathbf{Q}_{g_\pi}$ represents a permutation in S_K while for $\tilde{g}_\pi \in \mathcal{G}$ we have $\mathbf{Q}_{\tilde{g}_\pi} = -\mathbf{Q}_{g_\pi}$. Therefore, $\mathbf{Q}_g$ is orthogonal for every $g \in \mathcal{G}$. It follows that for every $\xi \in \Xi$ and every $g \in \mathcal{G}$ the information matrix of ξ^g satisfies $\mathbf{M}(\xi^g) = \mathbf{Q}_g \mathbf{M}(\xi) \mathbf{Q}_g^\top$. Hence, due to the orthogonality of $\mathbf{Q}_g$, the vectors $\boldsymbol{\lambda}(\xi^g)$ and $\boldsymbol{\lambda}(\xi)$ of eigenvalues of $\mathbf{M}(\xi^g)$ and $\mathbf{M}(\xi)$ coincide up to order. Since the Φ_q -criterion depends only on the eigenvalues of the information matrix, it follows that $\Phi_q(\xi^g) = \Phi_q(\xi)$ for every $\xi \in \Xi$ and every $g \in \mathcal{G}$. In other words, the Φ_q -criterion is invariant with respect to the group $\mathcal{G}$.

For every $\xi \in \Xi$ let

$$\bar{\xi} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \xi^g$$

be its symmetrization with respect to $\mathcal{G}$ (Schwabe 1996, p. 21). Together the concavity of Φ_q and the invariance with respect to $\mathcal{G}$ yield

$$\Phi_q(\bar{\xi}) \geq \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \Phi_q(\xi^g) = \Phi_q(\xi).$$

Since for any $h \in \mathcal{G}$ we have $\{h \circ g : g \in \mathcal{G}\} = \mathcal{G}$, the definition of $\bar{\xi}$ implies for the induced design $\bar{\xi}^h$ that $\bar{\xi}^h = \bar{\xi}$. Therefore, $\bar{\xi}$ is invariant with respect to $\mathcal{G}$. Thus, for every design $\xi \in \Xi$ there exists an invariant design $\bar{\xi} \in \Xi$ that is at least as good as ξ in terms of the Φ_q -criterion. In particular, this applies if $\xi \in \Xi'$.

Altogether we can conclude that for every $\eta \in \Xi$ there exists a design $\xi \in \Xi'$ with $\Phi_q(\xi) \geq \Phi_q(\eta)$ by Theorem 1 and that for the symmetrization $\bar{\xi}$ of this ξ we have $\Phi_q(\bar{\xi}) \geq \Phi_q(\xi)$ so that $\Phi_q(\bar{\xi}) \geq \Phi_q(\eta)$. In order to complete the proof, it remains to show that $\bar{\xi} \in \Xi^* \subset \Xi'$ for every $\xi \in \Xi'$.

Let $\xi \in \Xi'$, then $\xi(\mathcal{X}') = 1$. As $\mathcal{X}' = \mathcal{O}_{a,b} \cup \mathcal{O}_b$, it follows immediately from $g^{-1}(\mathcal{O}_{a,b}) = \mathcal{O}_{a,b}$ and $g^{-1}(\mathcal{O}_b) = \mathcal{O}_b$ that $g^{-1}(\mathcal{X}') = \mathcal{X}'$ for every $g \in \mathcal{G}$. For the symmetrization $\bar{\xi}$ of ξ this implies $\bar{\xi}(\mathcal{X}') = 1$ so that $\bar{\xi} \in \Xi'$. Next, we show how the symmetrization $\bar{\xi}$ of $\xi \in \Xi'$ meets the additional conditions for designs in Ξ^* .

To this end, consider for every $(\mathbf{s}, \mathbf{t}) \in \mathcal{X}$ the set $\mathcal{G}(\mathbf{s}, \mathbf{t}) = \{g(\mathbf{s}, \mathbf{t}) : g \in \mathcal{G}\}$, which is known as the orbit of $(\mathbf{s}, \mathbf{t})$ under $\mathcal{G}$. Since $\mathcal{G}$ is finite, $\mathcal{G}(\mathbf{s}, \mathbf{t})$ is finite too. Let $N(\mathbf{s}, \mathbf{t})$ denote the number of elements of $\mathcal{G}(\mathbf{s}, \mathbf{t})$. The points in $\mathcal{G}(\mathbf{s}, \mathbf{t})$ can be obtained by fixing an arbitrary point $(\tilde{\mathbf{s}}, \tilde{\mathbf{t}}) \in \mathcal{G}(\mathbf{s}, \mathbf{t})$ and applying $N(\mathbf{s}, \mathbf{t})$ transformations in $\mathcal{G}$ to $(\tilde{\mathbf{s}}, \tilde{\mathbf{t}})$. For technical convenience we express these transformations as inverse transformations $g_1^{-1}, \dots, g_{N(\mathbf{s}, \mathbf{t})}^{-1}$ of $g_1, \dots, g_{N(\mathbf{s}, \mathbf{t})} \in \mathcal{G}$. For every $(\tilde{\mathbf{s}}, \tilde{\mathbf{t}}) \in \mathcal{G}(\mathbf{s}, \mathbf{t})$ the orbit can then be represented as

$\mathcal{G}(\mathbf{s}, \mathbf{t}) = \{g_1^{-1}(\tilde{\mathbf{s}}, \tilde{\mathbf{t}}), \dots, g_{N(\mathbf{s}, \mathbf{t})}^{-1}(\tilde{\mathbf{s}}, \tilde{\mathbf{t}})\}$. For every design $\tilde{\xi} \in \Xi$ that is invariant with respect to $\mathcal{G}$ we have $\tilde{\xi}(\{g_i^{-1}(\tilde{\mathbf{s}}, \tilde{\mathbf{t}})\}) = \tilde{\xi}(g_i^{-1}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\})) = \tilde{\xi}^{g_i}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\}) = \tilde{\xi}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\})$ which implies that

$$\tilde{\xi}(\mathcal{G}(\mathbf{s}, \mathbf{t})) = \sum_{i=1}^{N(\mathbf{s}, \mathbf{t})} \tilde{\xi}(\{g_i^{-1}(\tilde{\mathbf{s}}, \tilde{\mathbf{t}})\}) = \sum_{i=1}^{N(\mathbf{s}, \mathbf{t})} \tilde{\xi}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\}) = N(\mathbf{s}, \mathbf{t})\tilde{\xi}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\}).$$

Since $(\tilde{\mathbf{s}}, \tilde{\mathbf{t}}) \in \mathcal{G}(\mathbf{s}, \mathbf{t})$ was fixed but arbitrary, this shows that the invariant design $\tilde{\xi}$ assigns the same weight $\tilde{\xi}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\}) = \frac{\tilde{\xi}(\mathcal{G}(\mathbf{s}, \mathbf{t}))}{N(\mathbf{s}, \mathbf{t})}$ to every pair $(\tilde{\mathbf{s}}, \tilde{\mathbf{t}}) \in \mathcal{G}(\mathbf{s}, \mathbf{t})$.

If $(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_{a,b}$, then $\mathcal{G}(\mathbf{s}, \mathbf{t}) = \mathcal{O}_{a,b}$. Likewise, for every $(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_b$ we have $\mathcal{G}(\mathbf{s}, \mathbf{t}) = \mathcal{O}_b$. So, if in addition to being invariant with respect to $\mathcal{G}$ the design $\tilde{\xi}$ belongs to Ξ' , it follows that $\tilde{\xi}(\mathcal{G}(\mathbf{s}, \mathbf{t})) = 0$ for all $(\mathbf{s}, \mathbf{t}) \notin \mathcal{X}'$. Moreover, for such a $\tilde{\xi}$ we have $\tilde{\xi}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\}) = \frac{w}{|\mathcal{O}_{a,b}|}$ for all $(\tilde{\mathbf{s}}, \tilde{\mathbf{t}}) \in \mathcal{O}_{a,b}$ and $\tilde{\xi}(\{\tilde{\mathbf{s}}, \tilde{\mathbf{t}}\}) = \frac{1-w}{|\mathcal{O}_b|}$ for all $(\tilde{\mathbf{s}}, \tilde{\mathbf{t}}) \in \mathcal{O}_b$, where $w = \tilde{\xi}(\mathcal{O}_{a,b})$. Therefore, $\tilde{\xi} \in \Xi^*$.

Finally, noting that for the symmetrization $\bar{\xi}$ of a design $\xi \in \Xi'$ it was shown that $\bar{\xi}$ is invariant with respect to $\mathcal{G}$ and that $\bar{\xi} \in \Xi'$, we conclude that $\bar{\xi} \in \Xi^*$. $\square$

Proof of Lemma 2 The information matrix of $\xi_w \in \Xi^*$ is

$$\mathbf{M}(\xi_w) = \int_{\mathcal{X}} \mathbf{F}\mathbf{F}^\top d\xi_w = \frac{w}{|\mathcal{O}_{a,b}|} \sum_{(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_{a,b}} \mathbf{F}(\mathbf{s}, \mathbf{t})\mathbf{F}(\mathbf{s}, \mathbf{t})^\top + \frac{1-w}{|\mathcal{O}_b|} \sum_{(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_b} \mathbf{F}(\mathbf{s}, \mathbf{t})\mathbf{F}(\mathbf{s}, \mathbf{t})^\top.$$

By using $\mathbf{F}(\mathbf{s}, \mathbf{t})\mathbf{F}(\mathbf{s}, \mathbf{t})^\top = \mathbf{s}\mathbf{s}^\top - \mathbf{t}\mathbf{s}^\top - \mathbf{s}\mathbf{t}^\top + \mathbf{t}\mathbf{t}^\top$ and the specific form of the pairs in $\mathcal{O}_{a,b}$ and $\mathcal{O}_b$ we get

$$\sum_{(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_{a,b}} \mathbf{F}(\mathbf{s}, \mathbf{t})\mathbf{F}(\mathbf{s}, \mathbf{t})^\top = 2(a^2 + b^2)(K-1)\mathbf{I}_K - 4ab(\mathbf{J}_K - \mathbf{I}_K)$$

and

$$\sum_{(\mathbf{s}, \mathbf{t}) \in \mathcal{O}_b} \mathbf{F}(\mathbf{s}, \mathbf{t})\mathbf{F}(\mathbf{s}, \mathbf{t})^\top = 2b^2(K-1)\mathbf{I}_K - 2b^2(\mathbf{J}_K - \mathbf{I}_K).$$

Because of $|\mathcal{O}_{a,b}| = 2K(K-1)$ and $|\mathcal{O}_b| = K(K-1)$ this gives

$$K(K-1)\mathbf{M}(\xi_w) = b^2w((c^2 + 1)(K-1)\mathbf{I}_K - 2c(\mathbf{J}_K - \mathbf{I}_K)) + 2b^2(1-w)(K\mathbf{I}_K - \mathbf{J}_K).$$

Rearranging terms yields (5).

Since $\mathbf{M}(\xi_w)$ is completely symmetric with main diagonal elements $\frac{b^2}{K}m_1(w)$ and off-diagonal elements $-\frac{2b^2}{K(K-1)}m_2(w)$, the formulae for the eigenvalues $\lambda_1(\xi_w)$ and $\lambda_2(\xi_w)$ and their multiplicities follow immediately from standard results (see, e.g., Rao 1973, p. 67). Finally, from $\lambda_1(\xi_w) \geq 0$ and $\lambda_2(\xi_w) - \lambda_1(\xi_w) = \frac{2b^2}{K-1}m_2(w) > 0$ for $w \in [0, 1]$ we obtain $0 \leq \lambda_1(\xi_w) < \lambda_2(\xi_w)$. $\square$

Proof of Theorem 3 For $q = -\infty$, the smallest eigenvalue of $\mathbf{M}(\xi_w)$ needs to be maximized. According to Lemma 2, this is the eigenvalue $\lambda_1(\xi_w)$ in (6). Obviously, $\lambda_1(\xi_{w-\infty}) \geq \lambda_1(\xi_w)$ for all $w \in [0, 1]$. Hence the design $\xi_{w-\infty}$ is $\Phi_{-\infty}$ -optimal in Ξ^* and Ξ .

Next, consider $q = 0$. Since $\lambda_1(\xi_w) = 0$ and $\Phi_0(\xi_w) = 0$ for $w = 0$, it suffices to consider $w > 0$. For every $\xi_w \in \Xi^*$ with $0 < w \leq 1$ the value of the Φ_0 -criterion can be expressed as

$$\Phi_0(\xi_w) = (\lambda_1(\xi_w)(\lambda_2(\xi_w))^{K-1})^{1/K} = \frac{b^2}{K(K-1)}(f(w))^{1/K},$$

where

$$f(w) = (1-c)^2(K-1)w(2K - \alpha(c, K)w)^{K-1}.$$

Let $w^* = \frac{2}{\alpha(c, K)}$. Consider f as a function with domain $D = [0, Kw^*]$. Then $f(0) = f(Kw^*) = 0$ and $f(w) > 0$ for every $0 < w < Kw^*$. Straightforward calculus shows that on D the function f attains a unique maximum in w^* and that f is strictly increasing on $[0, w^*]$. Hence, when f is restricted to $[0, 1] \subset D$, the function attains its maximum at $w_0 = \min(w^*, 1)$. It follows that for $\xi_w \in \Xi^*$ the value of $\Phi_0(\xi_w)$ is maximized for $w = w_0$ which demonstrates the Φ_0 -optimality of the design ξ_{w_0} .

Finally, consider $q = -1$. As for $q = 0$, we only need to consider designs $\xi_w \in \Xi^*$ with $0 < w \leq 1$. For these, the value of the Φ_{-1} -criterion can be written as

$$\Phi_{-1}(\xi_w) = \left(\frac{1}{K} \frac{\lambda_2(\xi_w) + (K-1)\lambda_1(\xi_w)}{\lambda_1(\xi_w)\lambda_2(\xi_w)} \right)^{-1} = \frac{b^2}{K-1} r(w),$$

where $r(w) = g(w)/h(w)$ with

$$g(w) = (1-c)^2(K-1)w(2K - \alpha(c, K)w)$$

and

$$h(w) = 2K + \left((1-c)^2(K-1)^2 - \alpha(c, K) \right) w.$$

Consider r , g , and h as functions whose domain is again $D = [0, Kw^*]$ with $w^* = \frac{2}{\alpha(c, K)}$. Then $g(0) = g(Kw^*) = 0$ and $g(w) > 0$ for every $w \in (0, Kw^*)$. Moreover, since $g(w)$ is quadratic in w , the function assumes a unique maximum in $Kw^*/2$ and $g(w)$ is strictly increasing for $w \in [0, Kw^*/2]$. The function h is linear in w with slope equal to zero for $c = \frac{K-3}{K-1}$ and $c = 1$. Thus, when h is constant, the design $\xi_{\tilde{w}^*} \in \Xi^*$ with $\tilde{w}^* = \min(Kw^*/2, 1)$ is Φ_{-1} -optimal in Ξ^* and Ξ . Note, that in this case $\tilde{w}^* = w_{-1}$.

For $0 < c < \frac{K-3}{K-1}$ the slope of h is positive and for $\frac{K-3}{K-1} < c < 1$ it is negative, which follows from the fact that for $K \geq 2$ the slope of h is quadratic in c where the coefficient of c^2 is positive. Nevertheless, $h(w) > 0$ for all $w \in D$. It follows that $r(0) = r(Kw^*) = 0$ and $r(w) > 0$ for very $0 < w < Kw^*$. Equating the derivative of $\ln r(w)$ for $0 < w < Kw^*$ to zero and solving the resulting equation for w is equivalent to determining the roots of another quadratic polynomial in w . The smaller of these roots is

$$z_1 = \frac{2K}{\alpha(c, K) + (1-c)(K-1)\sqrt{\alpha(c, K)}}.$$

Clearly $z_1 < Kw^*$, but $z_1 > 1$ is possible. In the expression for the larger root z_2 the plus in the denominator of z_1 is replaced with a minus so that $z_2 > Kw^*$. Therefore, on D the function r possesses a unique maximum at z_1 . Since $r(w)$ is strictly increasing for $w \leq z_1$, it follows that the design ξ_w with $w = w_{-1}$ is Φ_{-1} -optimal in Ξ^* and Ξ . $\square$

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